

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1**(Adv-2021-P1-Model) **Date: 22-04-2026**

Time: 09.00Am to 12.00Pm **GTA-26** **Max. Marks: 180**

22-04-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2021-P1)_GTA-26_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

Rankers Academy JEE

Name of the Student: _____ **H.T. NO:** | | | | | | | | |

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JEE-ADVANCE-2021-P1-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 10)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 11 – 16)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – IV(Q.N : 17 – 19)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 20 – 23)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 24 – 29)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 30 – 35)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – IV(Q.N : 36– 38)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 39 – 42)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 43 – 48)	Paragraph Questions with Numerical Value Answer Type	+2	0	6	12
Sec – III(Q.N : 49 – 54)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – IV(Q.N : 55 – 57)	Questions with Non-negative Integer Value Type	+4	0	3	12
Total				**19**	**60**

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PHYSICS **Max Marks: 60**

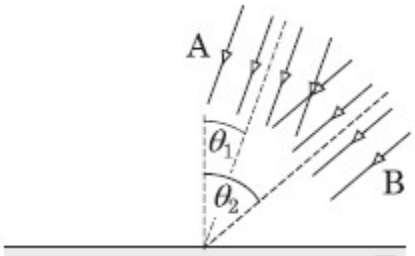
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)

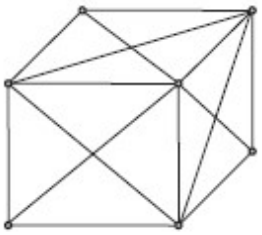
- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
- Full Marks: **+3** (If the answer is correct)
- Zero Marks: **0** (If the question is unanswered)
- Negative Marks: -1 (In all other cases)

1. Two identical beams A and B of plane coherent waves of the same intensity and wavelength λ fall on a plane screen. The directions of the beam propagations make angles $\theta_1 \text{ and } \theta_2$ with the normal to the screen and lie in the same plane as shown in the figure. Then the distance between adjacent interference fringes on the screen.



A) $\frac{\lambda}{\sin\theta_2 - \sin\theta_1}$ **B)** $\frac{\lambda}{\sin\theta_1} - \frac{\lambda}{\sin\theta_2}$ **C)** $\frac{\sin\theta_2 - \sin\theta_1}{\lambda}$ **D)** $\lambda(\sin\theta_2 - \sin\theta_1)$

2. In a well-known cube network, each edge contains a 24 Ω resistor and additional 12 Ω resistors along each face diagonal. These resistors are only connected electrically at the corners of the cube. For the sake of readability, in the figure only some of these branches without the resistors are shown. The equivalent resistance between two terminals connecting a face diagonal.



A) 7Ω **B)** 4.4Ω **C)** 6.4Ω **D)** 9Ω

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3. Current I flowing along edges of one face of a cube as shown in the Figure-I, produces magnetic field $\vec{B} = B_0\hat{j}$ at the centre of the cube. Consider another identical cube, where the current I flows along the path shown in the Figure-II. How much magnetic field exists at the centre of the second cube?

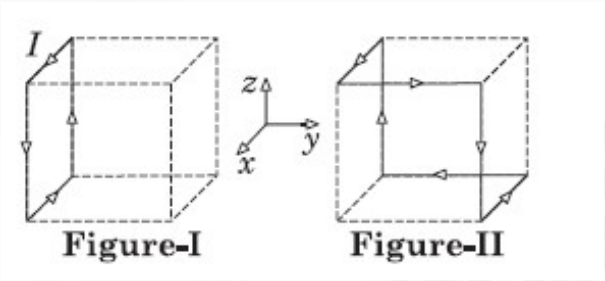
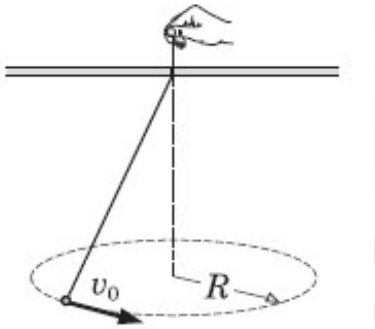


Figure-I Figure-II

A) Zero **B)** $B_0(\hat{i}+\hat{j}+\hat{k})$ **C)** $B_0(-\hat{i}+\hat{j}+\hat{k})$ **D)** $B_0(-\hat{i}+\hat{j}-\hat{k})$

4. A bead of mass m is attached to lower end of a light inextensible thread that passes through a small frictionless hole in a horizontal tabletop. The upper end of the thread is held motionless. The bead is pulled aside and projected horizontally to move on a circular path of radius R with a speed v_0 at a particular depth below the tabletop. Then the work done in slowly pulling the thread upwards to reduce the depth of circular path of the bead to half of its previous value.



A) $mv_0^2\left(\frac{g^2R^2}{v_0^4}+\sqrt{2}-1\right)$ **B)** $2mv_0^2\left(\frac{g^2R^2}{v_0^2}+\sqrt{2}-1\right)$

C) $\frac{mv_0^2}{2}\left(\frac{g^2R^2}{v_0^4}+\sqrt{2}-1\right)$ **D)** $mv_0^2\left(\frac{g^2R^2}{v_0^2}+\sqrt{2}-1\right)$

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SECTION-II
(PARAGRAPH WITH NUMERICAL VALUE TYPE)

- This section contains **THREE (03)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks:* +2** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

Question Stem for Question Nos. 5 and 6

Question Stem

A narrow beam of monochromatic light of wavelength λ emitted from a source of power P is propagating in the positive x-direction. After being reflected from a perfectly reflecting plane mirror of area vector $\vec{A} = A(-\hat{i} + \hat{j})$, the beam falls on a metal plate of surface area A placed parallel to the $x-z$ plane. The work function of the metal is ϕ and the photoelectric efficiency is η. Photoelectrons emitted from the metal plate are immediately removed from the vicinity of the plate.

5. Reflection from the mirror changes momentum of each photon by $\frac{h}{\lambda}\left(-x\hat{i} + \hat{j}\right)$ and exerts force on the mirror by the light beam is $\frac{P}{C}\left(\hat{i} - y\hat{j}\right)$. Then the value of $x + y$ is.

6. After a time interval Δt from the instant light falls on the metal plate. Then modules of electric field at a point very close to the metal plate and far away from its edges is $x\frac{\eta p \lambda e \Delta t}{\varepsilon_0 hCA}$ then x is

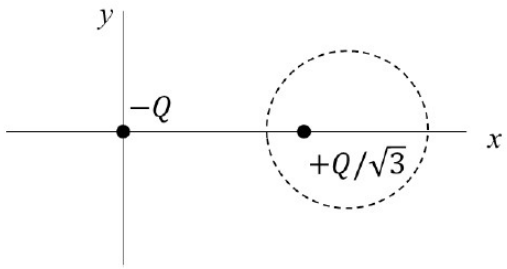
Question Stem for Question Nos. 7 and 8

Question Stem

Two point charges $-Q$ and $+Q/\sqrt{3}$ are placed in the $xy-$plane at the origin $(0,\ 0)$ and a point $(2,\ 0)$, respectively, as shown in the figure. This results in an equipotential circle of radius R and potential $V = 0$ in the $xy-$plane with its center at $(b,\ 0)$. All lengths are measured in meters.

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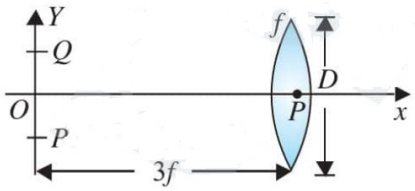


7. The value of R is ___ meter.
8. The value of b is ___ meter.

Question Stem for Question Nos. 9 and 10

Question Stem

An object kept near a convex lens of focal length f, executes SHM between P and Q according to the equation $y = A\sin\omega t, O$ being the mean position. Take x-axis as the principal axis of the lens and $A \ll D$ to answer the following questions



9. The amplitude of oscillation of the image is $xA\omega$, then x is

10. The velocity of the image when the object crosses the mean position and goes towards Q is $\frac{A\omega}{x}$, then x is

SECTION-III
(ONE OR MORE CORRECT ANSWER TYPE)

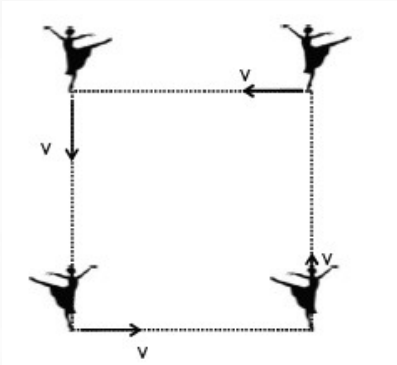
- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +4*** If only (all) the correct option(s) is (are) chosen;
- ***Partial Marks: +3*** If all the four options are correct but **ONLY** three options are chosen,
- ***Partial Marks: +2*** If three or more options are correct but **ONLY** two options are chosen, both of which are correct:
- ***Partial Marks: +1*** If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
- ***Zero Marks: 0*** If unanswered;
- ***Negative Marks: -2*** In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to the correct answer, then

Choosing ONLY (A), (B) and (D) will get +4 marks;
Choosing ONLY (A), will get +1 mark;
Choosing ONLY (B), will get +1 mark;
Choosing ONLY (D), will get +1 mark;
Choosing no option(s) (i.e. the question is unanswered) will get 0 marks and

Choosing any other option(s) will get -2 marks.

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11. Four dancers start their dance by forming a square with side $\ell = 9$ m. After that, they always move anticlockwise towards their neighbour with variable velocity $v = k\sqrt{x}$ where $k = 0.1\ \text{m}^{1/2}\text{sec}^{-1}$ and x is the distance of the closest neighbour in meter. Then choose the correct options



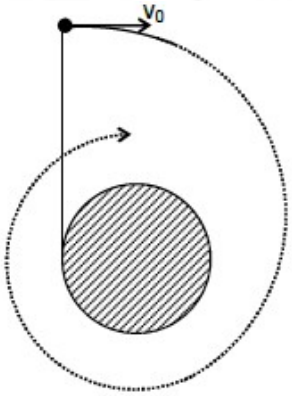
A) All the dancers will meet with each other at $t = 60$ sec after they start their dance.

B) Angular velocity of the line joining the adjacent dancers increases with time.

C) All the dancers will meet with each other at $t = 63$ sec after they start their dance.

D) Magnitude of acceleration of each dancer remains constant with time.

12. point object of mass $m = 2$ kg is connected to a cylinder of radius $R = 1$ m via a massless inextensible rope. At time $t = 0$, the object is moving with an initial velocity $v_0 = 5$ m/s perpendicular to the rope as shown in the figure. The rope has initial length $L_0 = 10$ m and having non-zero tension. Figure shows the top view and the motion of the object is taking place on the horizontal smooth surface and the cylinder is always fixed. The point object moves such that the rope wraps up around the cylinder. The rope will break when the tension in the rope exceeds 100 N .Then choose the correct option(s)



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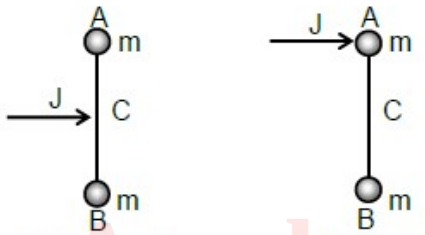
A) Kinetic energy of the object at the instant when the rope breaks is $25\ \text{J}$.

B) Angular momentum of the object about the axis of the cylinder at the instant when the rope breaks is less than $5\ \text{kg}\cdot\text{m}^2/\text{sec}$.

C) Angular momentum of the object about the axis of the cylinder at the instant when the rope breaks is equal to $5\ \text{kg}\cdot\text{m}^2/\text{sec}$.

D) Angular momentum of the object about the axis of the cylinder at the instant when the rope breaks is greater than $5\ \text{kg}\cdot\text{m}^2/\text{sec}$.

13. A dumbbell is formed by connecting two identical particles each of mass m with a massless rod. The system is placed on a horizontal smooth surface and impulse J is applied in two ways as shown in the figure as situation-(i) and situation-(ii). Then just after the application of impulse, choose the correct option(s).



A) Velocity of centre of mass with respect to ground frame in situation-(i) and situation-(ii) are same.

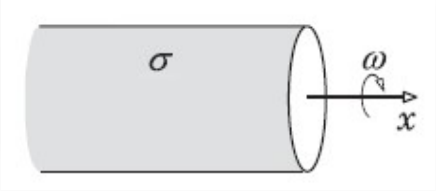
B) Kinetic energy of the dumbbell with respect to centre of mass frame in situation-(i) and situation-(ii) are same.

C) Angular momentum of the dumbbell with respect to centre of mass frame in situation-(i) and situation-(ii) are different.

D) Velocity of particle A in situation-(ii) is double of velocity of particle A in situation-(i) just after the application of impulse.

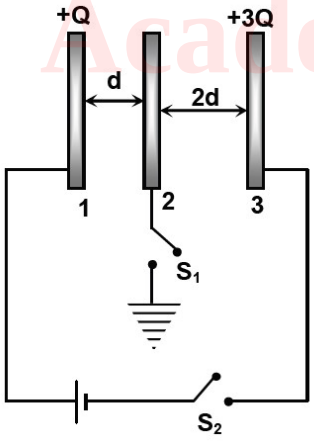
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14. A long thin wooden cylindrical pipe of radius R carrying a uniform surface charge density σ, is rotating about its axis with an angular velocity ω that increases slowly with time t as $\omega = kt$, where k is a positive constant. Which of the following statements is/are correct for the region inside the pipe?



A) Both the magnetic and the electric fields are uniform and constant.
B) The magnetic field is uniform but not constant and the electric field is constant but not uniform.
C) The magnetic field is constant but not uniform and the electric field is uniform but not constant.
D) Total energy density u due to both the fields varies with time as $u = a + bt^2$, where a and b are positive constants.

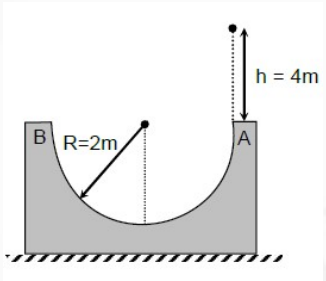
15. Three conducting sheets 1, 2 and 3 are placed parallel to each other such that separation between them is negligible. Sheet-1 and sheet-3 are given charges $+Q$ and $+3Q$ respectively. A cell of emf 1 V is to be connected between sheet 1 and 3. $(Take \frac{\epsilon_0 A}{d} = 30 \text{ mF}$ and $Q = 3 \text{ mC})$ where A is area of sheets. First switch S_1 is closed and then switch S_2 is closed.



A) When switch S_1 is closed charge flow through S_1 is zero.
B) When switch S_1 is closed charge flow through S_1 *is* 12 mC.
C) When S_2 is closed charge flown through S_2 is 2 mC.
D) When S_2 is closed charge flown through S_2 is 15 mC

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16. In the shown figure a block of mass 10 kg is placed on smooth horizontal surface. It has smooth hemispherical surface of radius R = 2m. A particle of mass 5 kg is released from rest from height of 4m above point A, such that it smoothly lands on spherical surface at point A. Neglect air drag .The correct option(s) are $\left(g = 10\ m/s^2\right)$



A) Particle will move up from end B upto height 4m

B) Particle will move up from end B upto height less then 4m

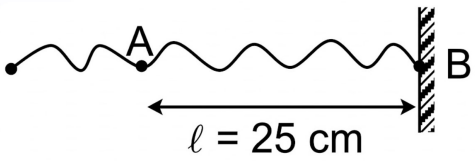
C) maximum vertical acceleration of the particle is $90\ m/s^2$

D) maximum vertical acceleration of particle is $40\ m/s^2$

SECTION-IV
(INTEGER ANSWER TYPE)

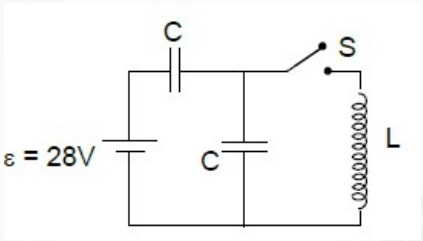
- This section contains **THREE (03)** question.
- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer the using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>
- ***Full Marks*** **: +4** If ONLY the correct integer is entered;
- ***Zero Marks*** **: 0** In all other cases.

17. A wave $y = 10\sin\left(200\pi t - \frac{4\pi}{100}x\right)$ (where x and y are in cm and t is in seconds) is travelling in a string in which the tension is $100\ \text{N}$. The wave is reflected from a fixed support and a stationary wave is produced. Assuming no loss of energy due to reflection and air drag, the energy in $25\ \text{cm}$ length of the string is $k\pi^2$ Joules. Find the value of k

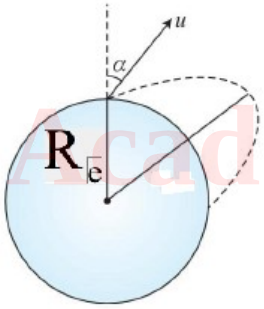


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18. In the shown figure two uncharged capacitors each of equal capacitance C = 2mF are connect to cell of emf 28V. The circuit was in this state from long time. Now switch S is closed with inductor with inductance L = 16mH. The maximum current in inductor in ampere will be



19. A projectile of mass m is fired from the surface of the earth at an angle $\alpha = 60°$ from the vertical. The initial speed v_0 is equal to $\sqrt{(GM_e / R_e)}$. Then the maximum height does the projectile rise above the earth surface is $\frac{R_e}{x}$ then x is ? (Neglect air resistance and the earth's rotation).



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CHEMISTRY **Max. Marks: 60**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)

- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
- Full Marks: **+3** (If the answer is correct)
- Zero Marks: **0** (If the question is unanswered)
- Negative Marks: -1 (In all other cases.

20. The correct statements is/are

1) CsCl crystal likely to show Schottky defect

2) AgBr exhibit both Schottky defect and Frenkel defect

3) Photosensitivity of AgBr is due to the presence of Frenkel defect in it

4) Crystals having F-centers are coloured and paramagnetic

A) Only 1 & 2 B) Only 3 & 4 C) All D) Only 1 & 3

21. The freezing point depression of a 0.10 M solution of formic acid is -0.2046°C. What is the equilibrium constant for the reaction at 298 K?

$HCOO^-(aq) + H_2O(\ell) \rightleftharpoons HCOOH(aq) + OH^-(aq)$

$\left(Given: K_f(H_2O) = 1.86K\ kgmol^{-1}, molarity = molality\right)$

(A) 1.11×10^{-3} **(B)** 9×10^{-12} **(C)** 9×10^{-13} **(D)** 1.1×10^{-11}

22. The emf of the cell $Ni\,|\,Ni^{2+}(0.1M)\,||\,Cu^{2+}(xM), H^+(0.01M)\,|\,Cu$ at 298 K is 0.5353 V. If the cathode solution is diluted with distilled water until the emf drops by 0.02955 V, the pH of the resulting cathode solution and the equilibrium constant for the reverse cell reaction are respectively: [Note: E^o of Cu^{2+}/Cu= 0.34V and E^o of Ni^{2+}/Ni= -0.25V].

A) $pH = 3,\ K_{rev} = 10^{\frac{-0.59\times2}{0.0591}}$ **B)** $pH = 4,\ K_{rev} = 10^{\frac{-0.59\times2}{0.0591}}$

C) $pH = 3,\ K_{rev} = 10^{\frac{0.59\times2}{0.0591}}$ **D)** $pH = 3.6,\ K_{rev} = 10^{\frac{-0.5353\times2}{0.0591}}$

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23. For a reaction $A \rightarrow B + C + ...Z$ (reactant and all products are gases), Which of the following is correct if the reaction follows 1st order kinetics. (P_i initial pressure, P_t total pressure at time eqbrm)

A) $k = \frac{2.303}{t}\frac{25P_i}{2P_i - P_t}$

B) $k = \frac{2.303}{t}\log\frac{24P_i}{2P_i - P_t}$

C) $k = \frac{2.303}{t}\log\frac{26P_i}{P_i - P_t}$

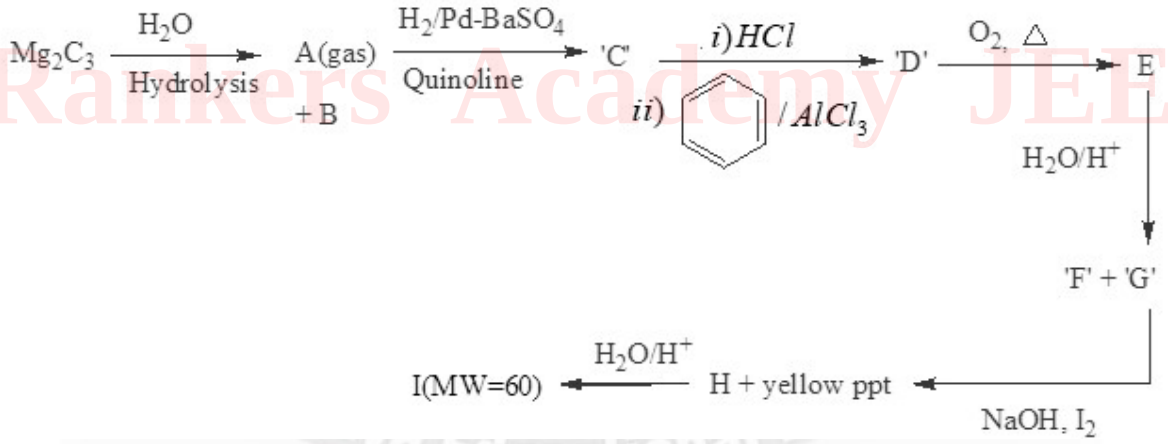
D) $k = \frac{2.303}{t}\log\frac{24P_i}{P_i - P_t}$

SECTION-II
(PARAGRAPH WITH NUMERICAL VALUE TYPE)

- This section contains **THREE (03)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>
- ***Full Marks*** **: +2** If ONLY the correct numerical value is entered at the designated place;
- ***Zero Marks*** **: 0** In all other cases.

Question Stem for Question Nos. 24 and 25

Question Stem



24. What is the degree of unsaturation in compound D?

25. What is the molecular weight of compound F?

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Question Stem for Question Nos. 26 and 27

Question Stem

It is observed that when acid and bases react, some energy is released due to neutralization reaction. While in case of strong acids and bases, the energy released is greater, in case of weaker acids or bases, energy released will be lesser. The difference is attributed to energy required for ionization. From the above information and data below answer the questions that follow :

ΔH neutralization SA/SB = – 57.5 kJ/equivalent;

ΔH ionization of NH_4OH = 10 kJ/mole;

ΔH ionization of $H_2C_2O_4$ = 20 kJ/mole.

26. 500 ml of 0.1 M H_2SO_4 solution reacts with 1L of 0.1 M NH_4OH solution if the base is 30% ionized in the given solution. The magnitude of heat energy released in this process is $x \times 10^{-2}$kJ, value of x is?

27. ΔH neutralization of $H_2C_2O_4$ with NaOH per mole of $H_2C_2O_4$ if acid is assumed to be completely unionized in the solution is |y| kJ/mole, what is the value of |y|?.

Question Stem for Question Nos. 28 and 29

Question Stem

The lowering of vapour pressure on adding a non-volatile solute may be measured by Ostwald- Walker method, In this method, dry air is first passed through a series of vessels having the solution, then through a series of vessels having pure solvent and finally through a vessel (normally U-tube) having the absorbent of the solvent. The masses of solution and pure solvent decreases due to removal of vapours of solvent in the flow of air and the mass of absorbent increases due to absorption of the vapour of solvent, in order to make the air dry. By measuring the changes in mass, we may determine the lowering of vapour pressure and hence, the composition of solution, form Raoul's law.

28. If the mass of absorbent increased by 0.24 gm and the mass of pure solvent (water) decreased by 0.01 gm ,then the mass percent of solute (glucose) in its aqueoues solution is :_____

29. If the experiment is performed with aq. $AlCl_3$ solution (α = 0.8) prepared by dissolving 1 mole $AlCl_3$ in 24 mole water and the decrease in mass of solution is found to be 0.1944 gm, then the increase in mass of absorbent should be 225×10^{-x} x=_____

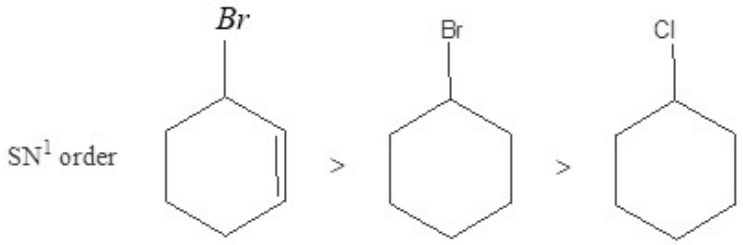
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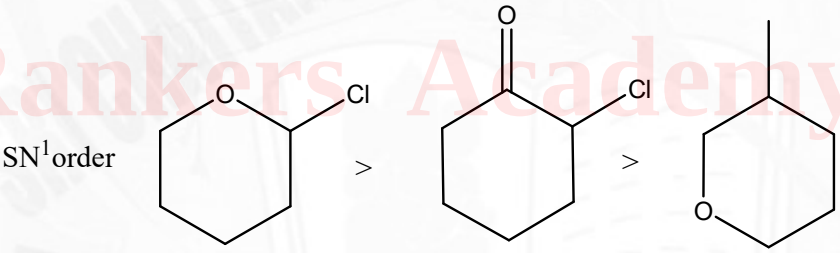
SECTION-III
(ONE OR MORE CORRECT ANSWER TYPE)

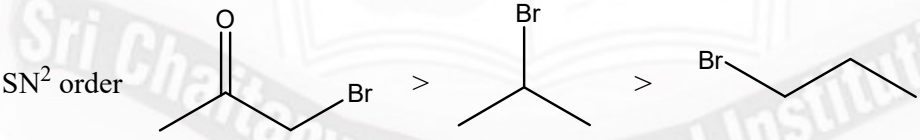
- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +4*** If only (all) the correct option(s) is (are) chosen;
- ***Partial Marks: +3*** If all the four options are correct but **ONLY** three options are chosen,
- ***Partial Marks: +2*** If three or more options are correct but **ONLY** two options are chosen, both of which are correct:
- ***Partial Marks: +1*** If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
- ***Zero Marks: 0*** If unanswered;
- ***Negative Marks: -2*** In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to the correct answer, then

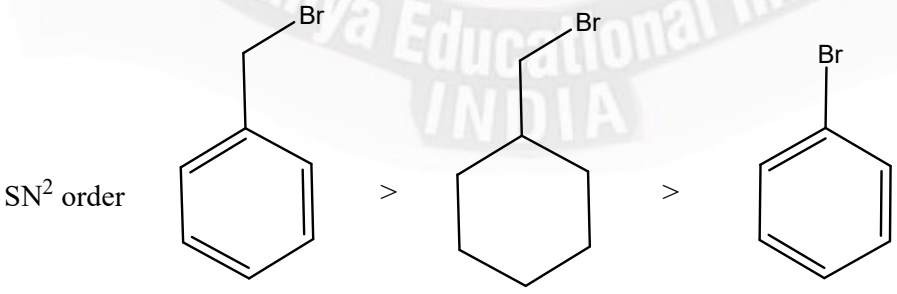
Choosing ONLY (A), (B) and (D) will get +4 marks;
Choosing ONLY (A), will get +1 mark;
Choosing ONLY (B), will get +1 mark;
Choosing ONLY (D), will get +1 mark;
Choosing no option(s) (i.e. the question is unanswered) will get 0 marks and
Choosing any other option(s) will get -2 marks.

30. Which of the following are correct order for the mentioned mechanism?

A) SN^1 order: 

B) SN^1 order: 

C) SN^2 order: 

D) SN^2 order: 

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31. When fructose treated with Tollen's reagent, silver mirror is formed due to reduction of $Ag^{\oplus}$ by :

A) fructose itself

B) glucose formed by isomerisation

C) mannose formed by isomerisation

D) galactose formed by isomerization

32. The correct statement(s) related to colloids is (are)

A) The process of precipitating colloidal sol by an electrolyte is called peptization.

B) Lyophilic solution is more viscous than lyophobic solution.

C) Surfactants form micelle above critical micelle concentration (CMC) and CMC depends on temperature.

D) Micelles are macromolecular colloids.

33. The diagram illustrates the energy change of a set of reaction



Which of the following statements are correct?

A) The enthalpy change for the reaction $P \rightarrow R$ *is* -33 kJ mol^{-1}

B) The enthalpy change for the transformation $R \rightarrow Q$ will be endothermic

C) The enthalpy change for the transformation $R \rightarrow Q$ will be exothermic

D) The enthalpy change for the transformation $S \rightarrow P$ *will be* $+42$ kJ mol^{-1}

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34. The correct statement(s) related to the metal extraction processes is(are)

A) In the extraction of iron in a blast furnace, limestone is added which first decomposes to CaO and CO_2, and the CaO then acts as a **reducing agent** to reduce Fe_2O_3 *to* Fe at high temperatures.

B) In the extraction of aluminium by Hall-Héroult process, pure Al_2O_3 has a very high melting point $(2050^{\circ}C)$, so it is dissolved in molten **cryolite** (Na_3AlF_6) to lower the melting point, and carbon anodes are consumed during the process because oxygen liberated at anode reacts with carbon to form CO_2 & *CO*

C) During the extraction of copper from copper pyrites $(CuFeS_2)$, the iron is removed as **iron silicate** $(FeSiO_3)$ slag, and the overall reaction during self-reduction gives blister copper which contains approximately 98% Cu with traces of SO_2 gas trapped inside — which is the reason it has a blistered appearance.

D) In the Bayer's process for purification of bauxite, the ore is dissolved in concentrated **NaOH** solution to form sodium aluminate, leaving impurities like Fe_2O_3 and TiO_2 undissolved. The filtrate pH is decreased to precipitate $\mathbf{Al(OH)_3}$, which is finally ignited to give Al_2O_3.

35. Aqueous solution of different salts give following reactions

$$P(aq) \xrightarrow{K_2CrO_4(aq)} \text{Yellow ppt} \xrightarrow[NaOH]{\text{Excess of}} \text{PPt will dissolve}$$

$$Q(aq) \xrightarrow{\text{dil NaOH(aq)}} \text{ppt is formed} \xrightarrow[\text{Ammonia}]{\text{Excess of}} \text{PPt will dissolve}$$

The salts in P and Q are respectively.

A) $Pb(NO_3)_2$, $CuSO_4$ B) $Pb(NO_3)_2$, $FeSO_4$

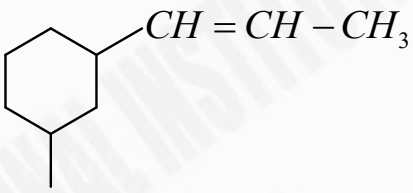
C) $BaCl_2$, $AlCl_3$ D) $BaCl_2$, $FeCl_3$

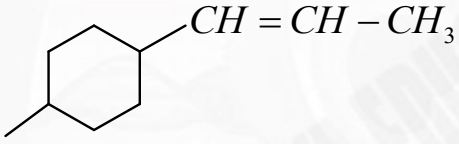
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SECTION-IV
(INTEGER ANSWER TYPE)

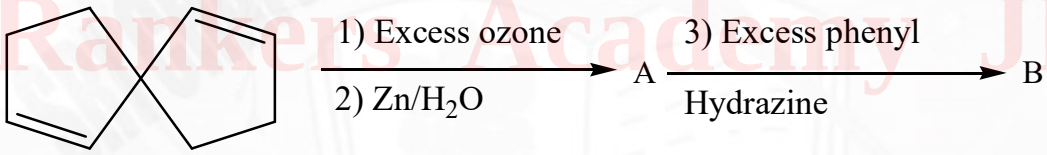
- This section contains **THREE (03)** question.
- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer the using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks*** **: +4** If ONLY the correct integer is entered;
- ***Zero Marks*** **: 0** In all other cases.

36. Sum of total number of stereo isomers for following

i) $H_3C-HC=C(CH=CHCH_3)_2$ ii) 

iii) 

37. In the reaction sequence given below, the total number of carbon atoms having sp^2 hybridization in the final product is _____(no intramolecular reaction takes place)



38. Number of cyclic rings in $[Co(edta)]^-$ is

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MATHEMATICS **Max. Marks: 60**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)

- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
- Full Marks: **+3** (If the answer is correct)
- Zero Marks: **0** (If the question is unanswered)

Negative Marks: -1 (In all other cases

39. The value of $\lim_{n\to\infty}\left(\lim_{m\to\infty}\left(\sum_{r=1}^{n}\left(\sum_{k=1}^{mr}\frac{mn^2}{(m^2n^2+k^2)(n^2+r^2)}\right)\right)\right)$ *is equal to*

A) $\frac{\pi^2}{16}$ **B)** $\frac{\pi^2}{32}$ **C)** $\frac{\pi}{32}$ **D)** $\frac{\pi^2}{8}$

40. Tangents are drawn to the parabola $y^2=4x$ from the point P (6, 5) to touch the parabola at Q and R. C_1 is a circle which touches the parabola at Q and C_2 is a circle which touches the parabola at R whose radii r_1 and r_2 respectively and both the circles C_1 and C_2 pass through the focus of the parabola. Then the value of $r_2^2-4r_1^2$ is equal to [where $r_2>r_1$]

A) 120 **B)** 100 **C)** 125 **D)** 250

41. Let $f(x)=\begin{cases}\dfrac{x^n\cos\left(\frac{1}{x}\right)}{\tan^m x}, & x\neq 0\\ 0, & x=0\end{cases}$ If f(x) is continuous but non-derivable at x=0 then

A) $n\le m$ **B)** $m<n\le m+1$ **C)** $n<2m$ **D)** $n<2m+1$

42. Mr. A travels from Delhi to Muscat with change of airplanes in Chennai, Dubai and Sharjah respectively. He has one piece of luggage. At each stop the luggage is transferred from one airplane to another. At the airport in Delhi there is a probability of $\frac{1}{20}$ that luggage is not placed in right plane. This probability is $\frac{1}{10}$ at airport in Chennai, $\frac{1}{5}$ at airport in Dubai and $\frac{1}{10}$ at airport in Sharjah, then If Mr. A luggage does not reach muscat with him, then the probability that it was lost at airport of Dubai is

A) $\frac{715}{1922}$ **B)** $\frac{855}{1922}$ **C)** $\frac{755}{1922}$ **D)** $\frac{705}{1922}$

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SECTION-II
(PARAGRAPH WITH NUMERICAL VALUE TYPE)

- This section contains **THREE (03)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +2*** If ONLY the correct numerical value is entered at the designated place;
- ***Zero Marks: 0*** In all other cases.

Question Stem for Question Nos. 43 and 44

Question Stem

In a tetrahedron $ABCD$, the face ABC is a regular (Equilateral) triangle and the face DBC is perpendicular to it. $\angle DAC = \frac{\pi}{3}, |\overline{AD}| = 6$ cm, & the angle between the lines $\overline{AD}$ & $\overline{BC}$ is equal to $\text{arc}\left(\cos\frac{1}{4}\right)$, if A is origin , $\overline{AD} = \bar{d}, \overline{AB} = \bar{b}, \overline{AC} = \bar{c}$

43. $|\vec{b}| + |\vec{c}| =$

44. The volume of tetrahedron is

Question Stem for Question Nos. 45 and 46

Question Stem

If $z_1, z_2, \ldots, z_8$ are the roots of the equation $(z^2 - 4z + 4)^4 = (9)^4$, such that $\arg z_1 < \arg z_2 < \ldots < \arg z_8$ (where $\arg z \in [0, 2\pi)$). Let z_k be a complex number satisfying the equation $|z_k - z_4| = |z_8 - z_k|$

45. Area of the quadrilateral formed by the complex numbers z_2, z_4, z_6, z_8 is

46. $\min \sum_{r=1}^{4} |z_{2r} - z_k|$ *is*

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Question Stem for Question Nos. 47 and 48

Question Stem

The equations of internal and external bisectors of $\angle B$ of a ΔABC are $x+y-1=0$ and $x-y+4=0$ respectively. The equations of internal and external bisectors of $\angle C$ are $3x-y+9=0$ and $x+3y+12=0$ respectively

47. If the circumcenter of ΔIBC is (a,b), then $2b-a$ is..... (Where I is the incentre of the triangle ABC)

48. **If** $\tan C=\frac{p}{q}$, where p, q are relatively prime positive integers then $p+q$ is equal to

SECTION-III

(ONE OR MORE CORRECT ANSWER TYPE)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks*** **: +4** If only (all) the correct option(s) is (are) chosen;
- ***Partial Marks*** **: +3** If all the four options are correct but **ONLY** three options are chosen,
- ***Partial Marks*** **: +2** If three or more options are correct but **ONLY** two options are chosen, both of which are correct:
- ***Partial Marks*** **: +1** If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
- ***Zero Marks*** **: 0** If unanswered;
- ***Negative Marks: -2*** In all other cases.
- For example, in a question, if (A), (B) and (D) are the **ONLY** three options corresponding to the correct answer, then

Choosing ONLY (A), (B) and (D) will get +4 marks;
Choosing ONLY (A), will get +1 mark;
Choosing ONLY (B), will get +1 mark;
Choosing ONLY (D), will get +1 mark;
Choosing no option(s) (i.e. the question is unanswered) will get 0 marks and
Choosing any other option(s) will get -2 marks.

49. If $A(z_1),B(z_2),C(z_3)$ are the vertices of a ΔABC such that

$$\begin{vmatrix} |z_1| & |z_2| & |z_3| \\ |z_2| & |z_3| & |z_1| \\ |z_3| & |z_1| & |z_2| \end{vmatrix}=0 \text{ and } z_1,z_2,z_3 \text{ are non-zero complex numbers},$$

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A) IF $AB = AC, then \frac{(z_1+z_2)(z_1+z_3)}{(z_2+z_3)^2}$ is purely real.

B) $Arg\left(\frac{z_2}{z_3}\right)^2 = Arg\left(\frac{z_2-z_1}{z_3-z_1}\right)$

C) The altitude from A meets the circumcircle at $D(z_4)$ then the reflection of D in the line BC is $\frac{Z_2Z_4+Z_3Z_4-Z_2Z_3}{Z_4}$

D) $Arg\left(\frac{z_1}{z_3}\right) = Arg\left(\frac{z_1-z_2}{z_3-z_2}\right)^2$

50. Consider the function

$f: R \to R, f(x) = \sqrt[3]{x+\sqrt{1+x^2}} + \sqrt[3]{x-\sqrt{1+x^2}}$ and $\sum_{n=1}^{n} f^{-1}(n) = n(n+1)\left[\frac{n(n+1)}{a} + \frac{3}{b}\right]$, then

A) $a+b=12$ **B)** $a-b=0$ **C)** $a=2b$ **D)** $a-b=4$

51. If $f:[0,1] \to R$ is a differentiable function and $f(0)=0, f(x)>0$ for all $x \in (0,1)$, then which of the following is necessarily true?

A) $\frac{f'(c)}{f(c)} = \frac{f'(1-c)}{f(1-c)}$ *for some* $c \in (0,1)$ **B)** $2\frac{f'(c)}{f(c)} = \frac{f'(1-c)}{f(1-c)}$ *for some* $c \in (0,1)$

C) $4\frac{f'(c)}{f(c)} = \frac{f'(1-c)}{f(1-c)}$ *for some* $c \in (0,1)$ **D)** $\frac{f'(c)}{f(c)} = 3\frac{f'(1-c)}{f(1-c)}$ *for somec* $\in (0,1)$

52. Let A and B be commutative square matrices of order three such that A is symmetric and B is skew-symmetric. If $C = (A+B)^T(A+B)(A-B)^{-1}$ (where X^T and $|X|$ denote transpose and determinant value of X respectively), then which of the following is true?

A) $A+B-C=0$ **B)** $A-B-C=0$ **C)** $|B|=0$ **D)** $2A-C=C^T$

53. If a function satisfies $(x-y)f(x+y)-(x+y)f(x-y)=2(x^2y-y^3)$ for all $x, y \in R$ *and* $f(1)=2$, then

A) $|f(x)|$ is non-differentiable at two points **B)** $|f(x)|$ is continuous everywhere

C) $f(-4)=12$ **D)** $f(x-1)=f(-x)\ \forall\ x \in R$

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54. A set of lines $x+y-2+\lambda_1(2x+y-3)=0$ represents incident rays on an ellipse $S=0$ and $2x+3y-23+\lambda_2(2x-y-3)=0$ represents the set of reflective rays from the ellipse where $\lambda_1,\lambda_2 \in R$. If $P(3,7)$ is a point on the ellipse. Then which of the following are correct.

A) Eccentricity of ellipse is $\frac{\sqrt{5}}{2\sqrt{2}+1}$

B) Radius of the director circle of the ellipse is $\frac{5}{2}$

C) Area of triangle formed by point P and two foci is 5

D) Eccentricity of ellipse is $\frac{\sqrt{5}}{\sqrt{2}+1}$

SECTION-IV
(INTEGER ANSWER TYPE)

- This section contains **THREE (03)** question.
- The answer to each question is a **NON-NEGATIVE INTEGER.**
- For each question, enter the correct integer corresponding to the answer the using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks*** **: +4** If ONLY the correct integer is entered;
- ***Zero Marks*** **: 0** In all other cases.

55. One of the roots of the equation $2000x^6+100x^5+10x^3+x-2=0$ is of the form $\frac{m\pm\sqrt{n}}{r}$, where m is a non zero integer and n & r are relatively prime natural numbers. The value of $m+n+r$ *is*

56. If the polynomial $f(x)=4x^4-ax^3+bx^2-cx+5$ *where* $a,b,c \in R$ has four positive real roots say r_1,r_2,r_3 *and* r_4, such that $\frac{r_1}{2}+\frac{r_2}{4}+\frac{r_3}{5}+\frac{r_4}{8}=1$. Find the value of '$a$'.

57. Let $f(n,r)=\frac{{}^nC_r}{{}^{r+3}C_r}$ and $g(n)=\sum_{r=0}^{n}(-1)^r f(n,r)$, then $\sum_{n=1}^{12}\frac{1}{g(n)}$ is

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